

India's Agricultural Crop Production Analysis (1997–2021)

Sprint Planning & Estimation

A Sprint is a fixed period or duration in which a team works to complete a set of tasks. An Epic is a big task or project that is too large to complete in one sprint. It is broken down into smaller tasks (stories) that can be completed over multiple sprints. A Story is a small task; it is part of an Epic. A Story Point is a number that represents how much effort a story takes to complete (usually in the form of the Fibonacci series).

Story Point Scale (1, 2, 3, 5)

- 1 – Very Easy task
- 2 – Normal task
- 3 – Moderate task
- 5 – Difficult task

Sprint 1

Data Collection (Epic 1)

USN	User Story / Task	Story Points
USN1	Collecting the agricultural crop production dataset (1997–2021)	2
USN2	Loading the dataset into the environment	1

Data Preparation (Epic 2)

USN	User Story / Task	Story Points
USN3	Handling missing values	3
USN4	Handling categorical values / creating derived fields (e.g., season, decade)	3
USN5	Handling inconsistency and outliers in production/yield data	3

Total Story Points in Sprint 1 = 2 + 1 + 3 + 3 + 3 = 12

Sprint 2

Data Visualization (Epic 3)

USN	User Story / Task	Story Points
USN6	KPIs – Total Production & Total Yield (2 KPI tiles)	2
USN7	Trend of Agricultural Yield (1997–2021) – Line Chart	2
USN8	State-wise Agricultural Production Distribution – India Map	3
USN9	Crop-wise Total Production Analysis – Horizontal Bar Chart	3
USN10	State-wise Cultivated Area Comparison – Grouped Bar Chart	3
USN11	Leading States by Agricultural Production – Bar Chart	2
USN12	Year-wise Total Agricultural Production (1997–2021) – Bar Chart	2
USN13	Production Distribution by Unit Type – Donut Chart	2
USN14	Season-wise Agricultural Production Share – Pie Chart	3

Dashboard (Epic 4)

USN	User Story / Task	Story Points
USN15	Developing an interactive Tableau dashboard	5

Story (Epic 5)

USN	User Story / Task	Story Points
USN16	Developing the data story / insights summary	5

Total Story Points in Sprint 2 = $2+2+3+3+3+2+2+2+3+5+5 = 32$

Total Story Points & Velocity

Sprint 1 = 12

Sprint 2 = 32

Velocity = Total Story Points Completed / Number of Sprints

Total Story Points = $12 + 32 = 44$

Number of Sprints = 2

Velocity = $44 / 2 = 22$ (Story Points per Sprint)

Your team's velocity is 22 Story Points per Sprint.